

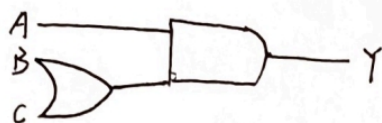
ch04

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1. Design circuit: $Y = A(B+C)$



2. Boolean Algebra to simplify.

$$(1) F_1 = \overline{AB + \bar{A}\bar{B}} (\bar{A}\bar{B} + CD)$$

$$= \overline{A+B} (\bar{A}\bar{B} + CD)$$

$$= \bar{A}\bar{B} (\bar{A}\bar{B} + CD)$$

$$= \bar{A}\bar{B}$$

$$(4) F_4 = (A + \bar{A}\bar{B}) (\overline{A + BC + \bar{C}})$$

$$= (A+B) (\overline{A+B+\bar{C}})$$

$$= (A+B) \bar{A}\bar{B}C$$

$$= (A+B) (\bar{A+B}) C$$

$$= 0$$

$$(2) \bar{F}_2 = \overline{A\bar{B}\bar{C} + AC + \bar{A}B\bar{C} + \bar{A}\bar{C}}$$

$$= \overline{A(\bar{B}+\bar{C}) + \bar{A}B\bar{C} + \bar{A}\bar{C}}$$

$$= \overline{A\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{C}}$$

$$= \overline{A \oplus B\bar{C}} + \bar{A}\bar{C}$$

$$= A \oplus B\bar{C} + \bar{A}\bar{C}$$

$$= A\bar{B}\bar{C} + \bar{A}(\bar{B}+\bar{C}) + \bar{A}\bar{C}$$

$$= A\bar{B}\bar{C} + \bar{A} = \bar{A} + B\bar{C}$$

$$(5) \bar{F}_5 = \overline{\bar{A}\bar{B} + \bar{A}CD + (B+\bar{C}+\bar{D})}$$

$$= \overline{(\bar{A}+\bar{B})(\bar{A}+\bar{C}+\bar{D})(B+\bar{C}+\bar{D})}$$

$$= \overline{(\bar{A}+\bar{B})(\bar{A}+\bar{C}+\bar{D})}$$

$$= \overline{\bar{A}\bar{C} + \bar{A}\bar{D} + \bar{A}B + B\bar{C} + B\bar{D}}$$

$$= \overline{\bar{A}\bar{C} + \bar{A}\bar{D} + \bar{A}B + B\bar{C}}$$

$$=$$

$$(5) F_5 = \overline{\bar{A}\bar{B} + \bar{A}CD (B+\bar{C}+\bar{D})}$$

$$= (\bar{A}+\bar{B})(\bar{A}+\bar{C}+\bar{D})(B+\bar{C}+\bar{D})$$

$$= \overline{(\bar{A}\bar{C} + \bar{A}\bar{D} + \bar{A}B + B\bar{C} + B\bar{D})(B+\bar{C}+\bar{D})}$$

$$= (\bar{A}+\bar{B})(\bar{A}+\bar{C}+\bar{D})$$

$$= \bar{A}\bar{C} + \bar{A}\bar{D} + B\bar{C} + \bar{A}B + B\bar{D}$$

$$= \bar{A}\bar{C} + \bar{A}B + B\bar{C} + \bar{A}B + \bar{A}\bar{D} + B\bar{D}$$

$$= \bar{A}\bar{C} + \bar{A}B + \bar{A}\bar{D}$$

$$(3) F_3 = (AB + \bar{A}\bar{B})(\bar{A}+\bar{B})\bar{A}\bar{B}$$

$$= (AB + \bar{A}\bar{B}) \cdot \bar{A}\bar{B} \cdot \bar{A}\bar{B}$$

$$= 0$$



$$(6) F_6 = (A+B)(A+\bar{A}\bar{B})C + \overline{\bar{A}(B+\bar{C})} + \bar{A}B + ABC$$

$$= (A+AB)C + A + \bar{B}C + \bar{A}B + ABC$$

$$= AC + ABC + A + \bar{B}C + \bar{A}B$$

$$= (A+AB)C + A + \bar{A}B + \bar{B}C$$

$$= A + \bar{A}B + \bar{B}C + AC$$

$$= A+B+C$$

3. Using K-Map to simplify

$$(1) F_1 = \sum m(3, 5, 6, 7)$$

A \ BC	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$$F_1 = BC + AC + AB$$

$$(2) F_2 = \sum m(4, 5, 6, 7, 8, 9, 10, 11, 12, 13)$$

AB \ CD	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	1	1	0	0
10	1	1	1	1

$$F_2 = \bar{D} + B\bar{C}$$

$$(3) F_3 = \sum m(2, 3, 6, 7, 10, 11, 12, 13)$$

AB \ CD	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	1	0	1	0
10	0	0	1	1

$$F_3 = AB\bar{C}\bar{D} + \bar{A}C + CD + \bar{B}C$$

$$(4) F_4 = \sum m(1, 3, 4, 5, 8, 9, 13, 15)$$

K-Map

AB \ CD	00	01	11	10
00	0	1	1	0
01	1	1	0	0
11	0	1	1	0
10	1	1	0	0

$$F_4 = \bar{C}\bar{D} + \bar{A}\bar{B}D + \bar{A}B\bar{C} + \bar{A}\bar{B}\bar{C} + ABD$$

$$(5) F_5 = \sum m(1, 3, 4, 6, 7, 9, 11, 12, 14, 15)$$

K-Map

AB \ CD	00	01	11	10
00	0	1	1	0
01	1	0	1	1
11	1	0	1	1
10	0	1	1	0

$$F_5 = BC + \bar{B}D + B\bar{D}$$

$$(6) F_6 = \sum m(0, 2, 4, 7, 8, 9, 12, 13, 14, 15)$$

AB \ CD	00	01	11	10
00	1	0	0	1
01	1	0	1	0
11	1	1	1	1
10	1	1	0	0

$$F_6 = \bar{C}\bar{D} + AB + A\bar{C}D + BCD + \bar{A}\bar{B}\bar{D}$$

